# MILITARY SPECIFICATION

# BREECHBLOCK-FUZE ASSEMBLY, ATOMIC WEAPON, TRAINING

## 1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of breechblock, fuze assembly, atomic weapon, training used on the 8-inch, M423 projectile.

## 2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein.

### STANDARDS

#### MILITARY

| | |
|---|---|
| MIL-STD-105 --- | Sampling Procedures and Tables for Inspection by Attributes |
| MIL-STD-1167 -- | Ammunition Data Card |
| MIL-STD-1168 -- | Lot Numbering of Ammunition |
| MIL-STD-1169 -- | Packaging, Packing and Marking for Shipment of Inert Ammunition Components. |

### DRAWINGS

#### PICATINNY ARSENAL

| | |
|---|---|
| 8787205 ------- | Breechblock, Fuze Assembly, Atomic Weapon, Training |
| 8787180 ------- | Block, Breech, Training |
| 8592949 ------- | Fuze, MT, M542 Assembly, Inert, Training |
| 8787183 ------- | Igniter, Training |
| 8794406 ------- | Ammunition Container Metal for Fuze |



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8794405 ------- Box, Packing, Ammunition for Fuze
ET8787205 ----- Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specified procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

## 3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The breechblock-fuze assembly, training shall comply with all requirements specified on the drawing (dwg) 8787205, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Workmanship.

3.3.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity or any similar characteristic.

3.3.2 Burrs. Parts shall be free from burrs.

3.3.3 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion or any embedded foreign material.

3.3.4 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

## 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Quality conformance inspection.

4.3.1 Lot formation.



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4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawings, specification and process changes not affecting safety performance, storage or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Tables I and II) and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.



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Table I
Major Characteristics

| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-72 | all | 140-141 | 124 | 256-259 | 177 | 541-549 | 230 |
| 73 | 72 | 142 | 125 | 260-262 | 178 | 550-559 | 231 |
| 74 | 73 | 143-144 | 126 | 263-265 | 179 | 560-569 | 232 |
| 75 | 74 | 145 | 127 | 266-268 | 180 | 570-579 | 233 |
| 76 | 75 | 146-147 | 128 | 269-271 | 181 | 580-590 | 234 |
| 77 | 76 | 148-149 | 129 | 272-275 | 182 | 591-601 | 235 |
| 78 | 77 | 150 | 130 | 276-278 | 183 | 602-613 | 236 |
| 79 | 78 | 151-152 | 131 | 279-282 | 184 | 614-625 | 237 |
| 80 | 79 | 153-154 | 132 | 283-285 | 185 | 626-637 | 238 |
| 81-82 | 80 | 155-156 | 133 | 286-289 | 186 | 638-649 | 239 |
| 83 | 8k | 157 | 134 | 290-293 | 187 | 650-662 | 240 |
| 84 | 82 | 158-159 | 135 | 294-296 | 188 | 663-676 | 241 |
| 85 | 83 | 160-161 | 136 | 297-300 | 189 | 677-690 | 242 |
| 86 | 84 | 162-163 | 137 | 301-304 | 190 | 691-704 | 243 |
| 87 | 85 | 164-165 | 138 | 305-308 | 191 | 705-719 | 244 |
| 88-89 | 86 | 166-167 | 139 | 309-312 | 192 | 720-735 | 245 |
| 90 | 87 | 168 | 140 | 313-316 | 193 | 736-751 | 246 |
| 91 | 88 | 169-170 | 141 | 317-320 | 194 | 752-768 | 247 |
| 92 | 89 | 171-172 | 142 | 321-324 | 195 | 769-785 | 248 |
| 93 | 90 | 173-174 | 143 | 325-329 | 196 | 786-804 | 249 |
| 94-95 | 91 | 175-176 | 144 | 330-333 | 197 | 805-823 | 250 |
| 96 | 92 | 177-178 | 145 | 334-338 | 198 | 824-842 | 251 |
| 97 | 93 | 179-180 | 146 | 339-342 | 199 | 843-863 | 252 |
| 98 | 94 | 181-182 | 147 | 343-347 | 200 | 864-884 | 253 |
| 99-100 | 95 | 183-184 | 148 | 348-352 | 201 | 885-907 | 254 |
| 101 | 96 | 185-187 | 149 | 353-356 | 202 | 908-930 | 255 |
| 102 | 97 | 188-189 | 150 | 357-361 | 203 | 931-955 | 256 |
| 103 | 98 | 190-191 | 151 | 362-366 | 204 | 956-981 | 257 |
| 104-105 | 99 | 192-193 | 152 | 367-372 | 205 | 982-1008 | 258 |
| 106 | 100 | 194-195 | 153 | 373-377 | 206 | 1009-1036 | 259 |
| 107 | 101 | 196-197 | 154 | 378-382 | 207 | 1037-1066 | 260 |
| 108-109 | 102 | 198-200 | 155 | 383-388 | 208 | 1067-1098 | 261 |
| 110 | 103 | 201-202 | 156 | 389-393 | 209 | 1099-1131 | 262 |
| 111 | 104 | 203-204 | 157 | 394-399 | 210 | 1132-1166 | 263 |
| 112-113 | 105 | 205-207 | 158 | 400-405 | 211 | 1167-1204 | 264 |
| 114 | 106 | 208-209 | 159 | 406-611 | 212 | 1205-1243 | 265 |
| 115 | 107 | 210-212 | 160 | 412-417 | 213 | 1244-1285 | 266 |
| 116-117 | 108 | 213-214 | 161 | 418-424 | 214 | 1286-1330 | 267 |
| 118 | 109 | 215-217 | 162 | 425-430 | 215 | 1331-1378 | 268 |
| 119 | 110 | 218-219 | 163 | 431-437 | 216 | 1379-1429 | 269 |
| 120-121 | 111 | 220-222 | 164 | 438-443 | 217 | 1430-1483 | 270 |
| 122 | 112 | 223-224 | 165 | 444-450 | 218 | 1484-1542 | 271 |
| 123-124 | 113 | 225-227 | 166 | 451-457 | 219 | 1543-1605 | 272 |
| 125 | 114 | 228-230 | 167 | 458-465 | 220 | 1606-1673 | 273 |
| 126-127 | 115 | 231-232 | 168 | 466-472 | 221 | 1674-1747 | 274 |
| 128 | 116 | 233-235 | 169 | 473-480 | 222 | 1748-1827 | 275 |
| 129-130 | 117 | 236-238 | 170 | 481-488 | 223 | 1828-1914 | 276 |
| 131 | 118 | 239-241 | 171 | 489-496 | 224 | 1915-2010 | 277 |
| 132-133 | 119 | 242-243 | 172 | 497-504 | 225 | 2011-2115 | 278 |
| 134 | 120 | 244-246 | 173 | 505-513 | 226 | 2116-2232 | 279 |
| 135-136 | 121 | 247-249 | 174 | 514-521 | 227 | 2233-2361 | 280 |
| 137 | 122 | 250-252 | 175 | 522-530 | 228 | over 2361 | 298 |
| 138-139 | 123 | 253-255 | 176 | 531-540 | 229 | | |



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Table II

Minor Characteristics

| Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size | Lot Size | Sample Size |
|---|---|---|---|---|---|---|---|
| 1-16 | all | 28-29 | 23 | 52-56 | 31 | 132-156 | 39 |
| 17 | 16 | 30-31 | 24 | 57-62 | 32 | 157-191 | 40 |
| 18 | 17 | 32-34 | 25 | 63-69 | 33 | 192-244 | 41 |
| 19-20 | 18 | 35-36 | 26 | 70-76 | 34 | 245-334 | 42 |
| 21 | 19 | 37-40 | 27 | 77-86 | 35 | 335-519 | 43 |
| 22-23 | 20 | 41-43 | 28 | 87-97 | 36 | 520-1120 | 44 |
| 24-25 | 21 | 44-47 | 29 | 98-112 | 37 | over 1120 | 45 |
| 26-27 | 22 | 48-51 | 30 | 113-131 | 38 | | |

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Igniter, training (see dwg. 8787183).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: None defined. | | |
| Major: | | |
| 101. | Major diameter of thread, max | Gage |
| 102. | Total length, max | Gage |
| 103. | Pitch diameter of thread, max | Gage |
| 104. | Outside diameter of tube section, max | Gage |
| Minor: | | |
| 201. | External diameter of thread pilot, max | Gage |
| 202. | Location of wrench holes in flange (functional) | Gage |
| 203. | Thickness of flange, max | Gage |
| 204. | Diameter of wrench holes, min | Gage |
| 205. | Chamfer incorrect or missing | Visual |
| 206. | Finish improper | Visual |
| 207. | Stamping missing, misleading or unidentifiable | Visual |
| 208. | Evidence of poor workmanship (see 3.3) | Visual |



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4.3.2.1.2 Breechblock-fuze assembly, atomic weapon, training (see dwg. 8787205).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: None defined. | | |
| Major: | | |
| 101. | Outside diameter, max. | Gage |
| 102. | Length, max. | Gage |
| 103. | Location of zero line incorrect. | Visual |
| Minor: | | |
| 201. | Marking missing, incorrect or illegible. | Visual |
| 202. | Protective finish improper. | Visual |

4.3.2.1.3 Block, breech, training (see dwg 8787180).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: None defined. | | |
| Major: | | |
| 101. | Pitch diameter of internal thread, minimum. | Gage |
| 102. | Major diameter of external thread, max. | Gage |
| 103. | Pitch diameter of external thread, max. | Gage |
| 104. | Outside diameter of front end. | Gage |
| Minor: | | |
| 201. | Pitch diameter of internal thread, max. | Gage |
| 202. | Pitch diameter of external thread, min. | Gage |
| 203. | Distance from rear face to external thread. | Gage |
| 204. | Distance from front face to external thread. | Gage |
| 205. | Depth of internal thread cavity, min. | Gage |
| 206. | Length of internal thread, min. | Gage |
| 207. | Total length. | Gage |
| 208. | Outside diameter of rear end. | Gage |
| 209. | Width of wrench slot. | Gage |



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210. Depth of wrench slot, min.....Gage
211. Inside diameter.....Gage
212. Location of wrench slots.....Gage
213. Location of threaded cavities.....Gage
214. Finish improper.....Visual
215. Stamping missing, mislocated, incorrect or unidentifiable.....Visual
216. Protective coating with bare spot.....Visual
217. Workmanship (see 3.3).....Visual

4.3.2.1.4 Preparation for delivery (see 5.1).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Preservation, packaging, packing and marking in accordance with MIL-P-1169 | Visual |
| Minor: | None defined. | |

4.3.2.1.5 Ammunition container, metal (see dwg F8794406).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: | None defined. | |
| Major: | | |
| 101. | Key missing or damaged | Visual |
| 102. | Container punctured, dented, damaged or deformed | Visual |
| Minor: | | |
| 201. | Contents loose | Manual |
| 202. | Marking missing, incorrect or illegible | Visual |



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4.3.2.1.6 Box, packing, ammunition for fuze (see dwg B8794405).

| Categories | Characteristics | Method of Inspection |
|---|---|---|
| Critical: None defined. | | |
| Major: | | |
| 101. | Hardware missing, broken, loose or improperly engaged | Manual/Visual |
| 102. | Rope handles insecure, damaged or missing | Manual/Visual |
| 103. | Strapping missing, broken, loose or improperly assembled | Manual/Visual |
| Minor: | | |
| 201. | Contents loose | Manual |
| 202. | Car seal missing, unsealed, or improperly positioned (when applicable) | Visual |
| 203. | Box damaged | Visual |
| 204. | Marking missing, incorrect or illegible | Visual |

4.3.3 Testing. Not applicable.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and minor characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment List (EL). EL's for assemblies are listed on the Equipment Tabulation, when required ET's or EL's, as applicable, shall be specified in the detail specifications (see 6.2). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.4 Test methods. Not applicable.



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## 5. PREPARATION FOR DELIVERY

5.1 Production items.

5.1.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1.1 Level A. The breechlock-fuze assembly, atomic weapon, training shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.1.2 Level C. The breechblock-fuze assembly, atomic weapon, training shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169(MU).

5.1.2 Packing.

5.1.2.1 Levels A and C. The breechblock-fuze assembly, atomic weapon, training shall be packed in accordance with the applicable instructions contained in MIL-STD-1169(MU).

5.1.3 Marking. Marking shall be in accordance with MIL-STD-1169(MU).

5.2 Repair parts.

5.2.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.2.1.1 Level A. The breechblock-fuze assembly, atomic weapon, training shall be packaged in accordance with dwg F8794406 and C8794405 using the specified moisture barrier.

5.2.1.2 Level C. The breechblock-fuze assembly, atomic weapon, training shall be packaged in accordance with dwg C8794405 except that the moisture barrier may be omitted.

5.2.2 Packing.

5.2.2.1 Level A and C. The breechblock-fuze assembly atomic weapon, training shall be packed in accordance with dwg F8794406 and C8794405.

5.2.3 Marking. Marking shall be as specified in accordance with dwg F8794406 and C8794405 as applicable.

## 6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.



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b. Production or repair parts.
c. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 or specification when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey, 07801, ATTN: SMUPA-QA-I.

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Project Number 1110-A-042



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# SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
MIL-B-48220 - Breechblock-Fuze Assembly, Atomic Weapon, Training

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☒ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.



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